HSR Human Point Follower: Gesture-Based Waypoint Navigation Using Real-Time Pose Estimation

Alex Morrow

Department of Computer Science

Abstract. This paper presents a complete system for interpreting human pointing gestures and autonomously navigating a mobile robot to the indicated target location. The implementation integrates real-time pose estimation using OpenPose with a robust finite state machine architecture that manages the full interaction lifecycle. A three-stage gesture validation pipeline distinguishes intentional pointing from ambient arm movements, while a face-to-hand vector projection method provides reliable 3D waypoint computation. The system is validated in NVIDIA Isaac Sim on a simulated HSR (Human Support Robot) platform, demonstrating successful end-to-end operation from gesture recognition to target arrival.

Keywords: Human-Robot Interaction · Gesture Recognition · OpenPose · Waypoint Navigation · ROS · Isaac Sim

1 Introduction

1.1 Problem Statement and Motivation

The ability for robots to understand and respond to natural human communication represents a fundamental challenge in human-robot interaction (HRI). This paper addresses a specific yet critical problem: *how can a robot reliably interpret a human pointing gesture and autonomously navigate to the indicated target location?* While pointing is among the most intuitive and universal forms of human non-verbal communication, translating this gesture into actionable robotic behavior requires solving several interconnected technical challenges.

Natural human-robot interaction is essential for deploying robots in real-world environments where non-expert users must communicate task objectives efficiently. Traditional interfaces requiring explicit programming, joystick control, or structured command languages create barriers to adoption and limit the practical utility of robotic systems in settings such as healthcare, domestic assistance, and collaborative manufacturing. Gesture-based interaction offers an intuitive alternative that leverages humans' existing communication skills without requiring specialized training.

This problem domain presents significant technical challenges that motivate our research. First, human pointing gestures exhibit substantial variation in pose,

arm extension, and body orientation across individuals and contexts. Second, computing accurate three-dimensional spatial information from two-dimensional image data requires sophisticated depth reasoning and geometric transformation. Third, the system must operate in real-time to provide responsive interaction, demanding efficient algorithms that balance accuracy with computational performance. Fourth, distinguishing intentional pointing gestures from incidental arm movements requires robust gesture recognition capable of filtering noise and validating user intent.

1.2 Project Goals

This project establishes a complete pipeline for gesture-based waypoint navigation with the following specific objectives:

- **Human Detection and Tracking:** Implement robust detection of human subjects within the robot's field of vision using the OpenPose BODY_25 model, with configurable confidence thresholds to filter noise and ensure reliable identification.
- **Pointing Gesture Recognition:** Develop a multi-stage validation system for recognizing pointing gestures, incorporating arm straightness verification, extension checking, and neutral arm filtering.
- **3D Waypoint Computation:** Calculate the target waypoint by projecting a face-to-hand pointing vector through three-dimensional space, leveraging combined RGB and depth sensor data.
- **Autonomous Navigation:** Execute navigation to the computed waypoint using dead-reckoning control, with state machine management to coordinate the complete interaction workflow.

1.3 Contributions

This work makes several key contributions to the field of gesture-based human-robot interaction:

- A comprehensive finite state machine implementation managing the complete interaction lifecycle through six distinct states (IDLE, SEARCHING, TRACKING, DETECTED, NAVIGATING, COMPLETED).
- A practical integration architecture connecting real-time pose estimation from OpenPose with the NVIDIA Isaac Sim environment and ROS middleware.
- An implementation of the face-to-hand pointing direction method, adapted from Azari et al.'s research [2], demonstrating its effectiveness for navigation waypoint computation.
- A novel three-stage validation pipeline combining arm straightness analysis, extension verification, and neutral position filtering to achieve robust pointing detection.

2 Background and Related Work

2.1 OpenPose

OpenPose is an open-source, real-time multi-person keypoint detection library developed by Carnegie Mellon University's Perceptual Computing Lab [1]. The system represents a significant advancement in computer vision, providing the capability to detect and track human body poses from standard RGB camera input. At its core, OpenPose detects body, hand, facial, and foot keypoints from RGB images, outputting a structured representation of human skeletal configuration.

The system operates by processing input images through a multi-stage convolutional neural network that simultaneously predicts confidence maps for body part locations and Part Affinity Fields (PAFs) that encode the degree of association between parts. These predictions are then parsed using a greedy algorithm to assemble detected parts into coherent skeletons for multiple individuals in the frame.

For this project, we selected the BODY_25 model as the primary data source for pose estimation. The BODY_25 model provides positions for 25 anchor points on the human skeleton, including key joints such as the nose, neck, shoulders, elbows, wrists, hips, knees, and ankles. Figure 1 illustrates the keypoint structure of the BODY_25 model. To optimize for real-time performance within the simulation environment, we configured OpenPose to operate at a reduced input resolution of 320×176 pixels, achieving approximately 15–30 frames per second.

2.2 Theoretical Foundation of Pointing Recognition

The theoretical foundation for our pointing gesture recognition approach draws significantly from the work of Azari, Lim, and Vaughan [2]. The central finding of their work, and the theoretical cornerstone of our implementation, is that the **face-to-hand vector best captures human pointing intention** when compared to alternative geometric models such as forearm direction or shoulder-to-hand vectors.

Figure 2 illustrates the face-to-hand pointing vector methodology. The pointing direction is computed from the face centroid to the wrist position, then projected through 3D space to find the intersection with the ground plane. This approach leverages the natural tendency of humans to align their gaze with their pointing direction.

Both approaches fundamentally rely on a combination of RGB and depth data to compute pointing direction in three-dimensional space. The RGB stream provides the visual information necessary for detecting and localizing human keypoints, while the depth stream enables projection of the 2D pointing vector into 3D world coordinates.

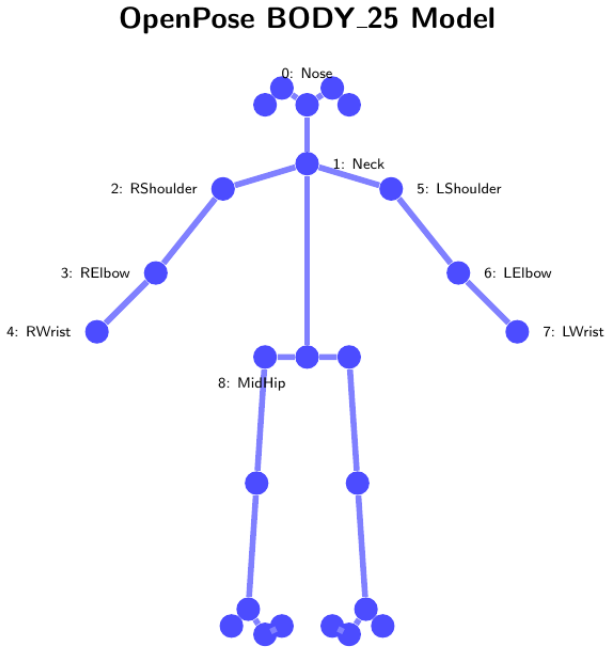

Fig. 1. The OpenPose BODY_25 keypoint model showing the 25 skeletal anchor points used for pose estimation. Key points relevant for pointing detection include the nose (0), neck (1), shoulders (2, 5), elbows (3, 6), and wrists (4, 7).

3 System Architecture and Components

3.1 Simulation and Hardware Platform

The system is developed for the HSR (Human Support Robot), a mobile manipulation platform designed by Toyota for assistive applications in home and healthcare environments. The HSR features an omnidirectional base, a single articulated arm, and a sensor suite including RGB-D cameras mounted on a pan-tilt head unit.

Development and testing were conducted within NVIDIA Isaac Sim, a physics-enabled, photorealistic simulation environment built on the Omniverse platform. Isaac Sim provides accurate physics simulation, realistic rendering, and native ROS integration, enabling rapid prototyping of robotic applications before deployment on physical hardware.

3.2 ROS Integration and Data Flow

The system architecture follows a modular ROS-based design, with distinct nodes responsible for specific functional components. Data flows through the system

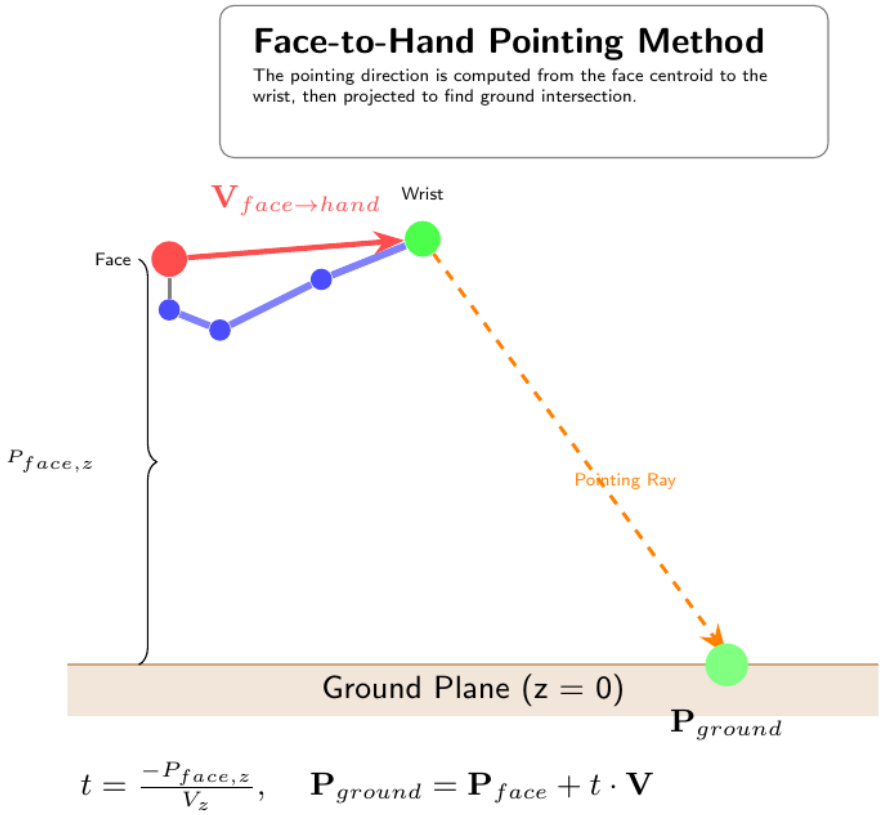

Fig. 2. The face-to-hand pointing vector method. The vector $\mathbf{V}$ from face to wrist defines the pointing direction, which is then projected to find the ground intersection point $\mathbf{P}_{ground}$ using ray-plane intersection.

in a pipeline fashion, from raw sensor acquisition through pose estimation to high-level control decisions. Figure 3 illustrates the overall system architecture.

The following components constitute the primary data flow:

- **HSR Robot Sensors:** The HSR provides RGB image data via the `/hsrb/head_rgbd_sensor/rgb/image_rect_color` topic and aligned depth data via the corresponding depth topic.
- **OpenPose Bridge:** This node subscribes to the RGB image stream and processes each frame through the OpenPose library, publishing visualization images and `PoseArray` messages containing 2D pixel coordinates.
- **Human Point Follower:** This node serves as the core control logic, consuming sensor data and OpenPose outputs to implement the complete HRI behavior.

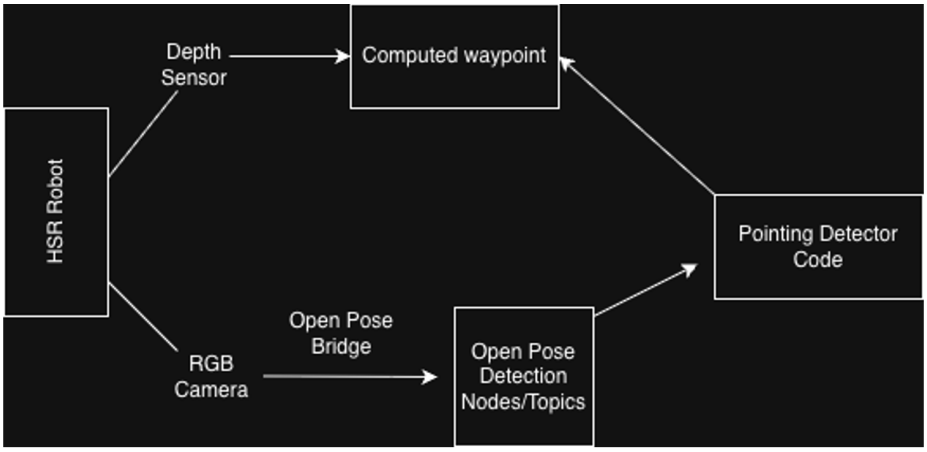

Fig. 3. System architecture showing data flow from HSR robot sensors through OpenPose detection to the pointing detector and waypoint computation modules.

Table 1. State Machine States, Behaviors, and Exit Conditions

State	Behavior	Exit Condition
IDLE	Wait for start signal	/start = True
SEARCHING	Rotate base 360°	Human detected (5 frames)
TRACKING	Keep human centered	Pointing gesture detected
DETECTED	Validate 3-second hold	Hold confirmed
NAVIGATING	Dead-reckoning to target	Reached destination
COMPLETED	Wait for reset signal	Message on /start topic

3.3 The Unified State Machine

The central control logic is implemented as a unified finite state machine that coordinates all aspects of the human-robot interaction task. Table 1 presents the six states comprising the interaction workflow.

4 Implementation Details

4.1 Pointing Gesture Recognition (3-Stage Validation)

The system validates a pointing gesture using three geometric checks:

Stage 1 – Arm Straightness Check: The angle between the upper arm vector (shoulder → elbow) and the forearm vector (elbow → wrist) is calculated in 2D image coordinates. The gesture is rejected if the angle deviates by more than 35°.

Stage 2 – Arm Extension Check: The ratio of the distance from the shoulder to the wrist to the maximum possible arm length is calculated. A ratio greater than 0.5 is required.

Stage 3 – Neutral Arm Filter: To distinguish an outstretched point from a relaxed, hanging arm, the filter rejects the gesture if the wrist's Y-pixel coordinate is significantly below the elbow's Y-pixel coordinate (threshold of 30 pixels).

4.2 3D Waypoint Computation

Upon successful hold confirmation, the system calculates the 3D ground target using the face-to-hand vector method. The pointing vector is defined as:

$$\mathbf{V} = \mathbf{P}_{\text{wrist}} - \mathbf{P}_{\text{face}} \tag{1}$$

Both the origin $\mathbf{P}_{\text{face}}$ and the vector $\mathbf{V}$ are transformed from the camera frame to the robot's reference frame using `tf2_ros`. The ground intersection point $\mathbf{P}_{\text{ground}}$ is calculated using the parametric line equation for the ground plane ($z = 0$):

$$t = \frac{-P_{\text{face},z}}{V_z}, \quad \mathbf{P}_{\text{ground}} = \mathbf{P}_{\text{face}} + t \cdot \mathbf{V} \tag{2}$$

4.3 Navigation (Dead Reckoning)

The robot transitions to NAVIGATING with the final target point. The system employs time-based dead reckoning with two phases:

Phase 0 – Rotation: The robot calculates the required rotation angle $\theta = \arctan 2(y, x)$ and rotates for duration $t_{\text{rot}} = |\theta|/\omega$.

Phase 1 – Driving: The robot drives straight for distance d with drive time $t_{\text{drive}} = d/v$.

5 Results, Limitations, and Conclusion

5.1 Results and Demonstration

The complete demonstration pipeline was successfully executed in the Isaac Sim environment, validating the end-to-end functionality of the gesture-based navigation system. Figure 4 shows the system in operation.

System accuracy was evaluated using RViz visualization markers displaying three key reference points: the detected human position, the ground truth target location, and the robot's computed waypoint. In controlled trials, the computed waypoint consistently fell within 0.3–0.5 meters of the ground truth marker at distances of 2–4 meters from the robot.

5.2 Limitations

Simulation Environment Constraints: The most significant limitation encountered was the lack of programmatic skeletal control in NVIDIA Omniverse's default human character models. This required development of a custom HumanArmController workaround.

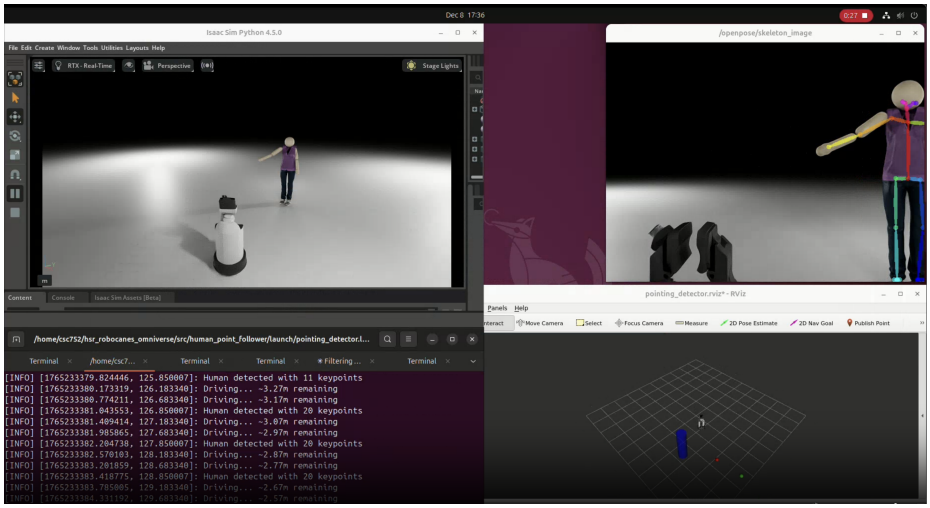

Fig. 4. System demonstration showing (left) Isaac Sim environment with HSR robot and human pointing, (top right) OpenPose skeleton detection overlay, (bottom right) RViz visualization with robot model and waypoint markers, and (bottom) terminal output showing human detection and navigation status.

Depth Sensing Challenges: Accurate depth sampling on small targets such as the wrist presented persistent challenges due to noise and interpolation artifacts.

Navigation Limitations: The time-based dead-reckoning navigation approach suffers from cumulative error that increases with travel distance.

5.3 Future Work

Several promising directions exist for extending this work. The most immediate enhancement would be replacing the dead-reckoning navigation with the ROS `move_base` action server. The complete pipeline could also serve as the basis for a Reinforcement Learning (RL) environment for training agents on complex HRI tasks.

5.4 Conclusion

This paper presented a complete system for interpreting human pointing gestures and navigating a mobile robot to the indicated target location. The implementation successfully integrates real-time pose estimation using OpenPose with a robust state machine architecture. The three-stage gesture validation pipeline effectively distinguishes intentional pointing from ambient arm movements, while the face-to-hand vector projection method provides reliable 3D waypoint computation.

References

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